

# From Proxy Features to Fair Decisions: Investigating Algorithmic Bias and Threshold Correction in NHS Mental Health AI Triage

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## Abstract

AI-assisted NHS mental health triage systems, such as CHRONOSIG, demonstrate the potential of machine learning to support clinical decision-making at scale. However, their use raises important questions regarding whether uniform decision thresholds can guarantee equitable outcomes across demographic groups. This project investigates this challenge and proposes a mathematically rigorous framework for navigating the resulting fairness trade-offs.

Drawing on Bayesian decision theory and Chouldechova’s (2017) impossibility result, I demonstrate that when demographic groups have different base rates of genuine clinical need, no uniform threshold can simultaneously satisfy calibration, equalised odds, and demographic parity. This challenge is particularly relevant in healthcare contexts, where disparities in data representation and access to care may contribute to differing base rates between demographic groups. These limitations arise from inherent conflicts between fairness objectives rather than from issues in model implementation.

I implemented a simulation using synthetic patient data informed by NHS statistics, demonstrating that under a baseline Random Forest classifier with a uniform threshold  $\tau = 0.4$ , Group A (a simulated underrepresented demographic group) experiences a false negative rate of 64.7%, compared with 6.7% for Group B. SHAP analysis identifies referral letter length, a documentation-derived feature, as a potential proxy feature associated with reduced referral predictions for 94.8% of Group A samples.

I propose a two-stage bias reduction framework. Stage 1 applies hyperparameter optimisation, comparing exhaustive Grid Search with Bayesian Optimisation using a Gaussian Process surrogate model, to investigate whether model tuning can reduce reliance on proxy variables by up to 87%. Stage 2 applies group-specific ROC thresholds derived by minimising group-specific cost-benefit ratios under a Bayesian loss function to manage fairness trade-offs between demographic groups. The framework is evaluated across Random Forest, XGBoost, and LightGBM architectures, demonstrating that the proposed approach is applicable across multiple

model families.

The framework is extended to Child and Adolescent Mental Health Services using synthetic patient data incorporating features derived from constructs measured by the Strengths and Difficulties Questionnaire (SDQ) and Revised Child Anxiety and Depression Scale (RCADS), demonstrating applicability across different NHS mental health settings.

The project’s contribution is to provide a practical framework for managing the fairness trade-offs identified by Chouldechova’s impossibility result. By making these trade-offs mathematically explicit and measurable, the framework supports informed and defensible threshold selection in NHS mental health AI triage.

**Keywords:** algorithmic fairness, NHS mental health triage, CHRONOSIG, Bayesian decision theory, ROC threshold optimisation, Chouldechova impossibility, equalised odds, CAMHS.

## 1 Introduction

Around 400,000 referrals are made monthly to NHS secondary mental health services, at a time when 1 in 5 children and young people in England are living with a probable mental health disorder. A substantial proportion of communication between patients and NHS mental health services is recorded as written clinical narratives, creating large volumes of documentation that can be time-consuming for clinicians to review. Machine learning systems have therefore been proposed as tools to support clinical decision-making and improve the efficiency of referral triage. One example is CHRONOSIG, which uses clinical information from referral documentation to assist multidisciplinary teams in prioritising patient pathways.

CHRONOSIG extracts information from referral documentation and available Electronic Health Records, transforming a patient’s medical history, symptoms, and presenting difficulties into a longitudinal signature. The system learns associations between these representations and historical triage decisions, allowing new patients to be compared with similar historical profiles to support clinical decision-making. Such systems are designed to assist rather than replace multidisciplinary clinical judgement, with final decisions remaining under human oversight.

Despite its potential, AI-assisted triage systems raise significant fairness concerns. As with any ML model, bias can arise from non-representative historical training data, potentially leading to unequal outcomes across demographic groups. This concern was also highlighted by patients themselves: in a focus group conducted by the CHRONOSIG research team in September 2024, participants independently raised concerns about non-representative training data, with one participant summarising the risk as “garbage in, garbage out” [Smith et al., 2026].

This project investigates whether uniform decision thresholds used in AI-assisted mental health triage systems, motivated by examples such as CHRONOSIG and

extended to CAMHS-inspired settings, can achieve equitable outcomes across demographic groups. Drawing on Bayesian decision theory and Chouldechova’s (2017) impossibility result, I examine the trade-offs that arise when demographic groups differ in base rates and predictive performance. I then propose a threshold optimisation framework that makes these fairness considerations explicit through group-specific decision boundaries and cost-sensitive analysis. The effectiveness of this approach is evaluated using synthetic NHS-inspired simulations, including CAMHS-inspired features informed by the Strengths and Difficulties Questionnaire (SDQ) and Revised Child Anxiety and Depression Scale (RCADS).

## 2 Background

### 2.1 Algorithmic Fairness Criteria

Algorithmic fairness research has introduced several criteria for assessing whether predictive models produce equitable outcomes across demographic groups. However, these criteria can conflict with one another, particularly when demographic groups have different underlying base rates. Let  $G \in \{A, B\}$  represent a binary demographic group,  $T \in \{0, 1\}$  represent the true clinical outcome (where  $T = 1$  indicates genuine need for secondary care), and  $D \in \{0, 1\}$  represent the final binary decision made after applying a classification threshold to the model output.

The simplest criterion, *demographic parity*, requires the probability of receiving a positive decision to be equal across demographic groups:

$$P(D = 1 \mid G = A) = P(D = 1 \mid G = B) \quad (1)$$

However, demographic parity does not account for differences in underlying clinical need and may therefore result in unequal predictive performance across groups.

A separate criterion is *equalised odds* [Hardt et al., 2016], which requires prediction performance to be independent of group membership conditional on the true outcome ( $D \perp G \mid T$ ). This requires equal true positive rates (TPR) and false positive rates (FPR) across groups:

$$P(D = 1 \mid T = 1, G = A) = P(D = 1 \mid T = 1, G = B) \quad (2)$$

$$P(D = 1 \mid T = 0, G = A) = P(D = 1 \mid T = 0, G = B) \quad (3)$$

Equalised odds therefore ensures that individuals with the same true outcome have equal probabilities of receiving a positive decision, regardless of demographic group.

Predictive parity, also referred to as calibration, is another fairness criterion that requires the positive predictive value (PPV) to be equal across demographic groups. This means that among individuals receiving a positive decision, the proportion with genuine clinical need should be consistent across groups:

$$P(T = 1 \mid D = 1, G = A) = P(T = 1 \mid D = 1, G = B) \quad (4)$$

Chouldechova formalised an impossibility result demonstrating that when the base rate of the true condition, defined as the prevalence  $P_G = P(T = 1 \mid G)$ , differs between demographic groups ( $P_A \neq P_B$ ), calibration, equalised odds, and demographic parity cannot all be satisfied simultaneously by a non-trivial classifier. Since healthcare datasets may contain differing base rates due to differences in representation, access to care, and historical processes, this result highlights the inherent trade-offs involved in selecting a single uniform decision threshold for AI-assisted triage.

## 2.2 Historical Bias and Proxy Variables in Healthcare AI

A key challenge in AI-assisted patient triage is the potential for bias arising from incomplete or non-representative clinical data. In written referrals, differences in documentation style, detail, and symptom description may result in unequal representation of patient needs across demographic groups. If these patterns are present in training data, models may learn associations that disadvantage certain populations.

This issue has been demonstrated in healthcare algorithms by Obermeyer et al. (2019), who analysed a widely used commercial algorithm designed to identify patients requiring additional healthcare support. The study found that, among patients assigned the same risk score, Black patients had substantially greater underlying illness burden than White patients. The algorithm therefore underestimated the healthcare needs of Black patients due to its reliance on healthcare expenditure as a proxy for illness severity. Because historical healthcare spending is influenced by unequal access to care, it can introduce systematic disparities into predictive models.

In reality, Black patients faced greater structural barriers to healthcare, including differences in income, insurance coverage, and access to services. As a result, healthcare expenditure was lower among some Black patients despite greater underlying health needs. The algorithm therefore learned an association between lower healthcare spending and lower clinical need, using resource utilisation as a proxy for illness severity.

By confusing historical patterns of healthcare utilisation with true clinical need, the training data can encode existing inequalities. Consequently, observed base rates in healthcare datasets may differ between demographic groups ( $P_A \neq P_B$ ), creating the conditions described by Chouldechova’s impossibility result, where a single uniform threshold cannot simultaneously satisfy all fairness objectives.

This example demonstrates that healthcare AI systems may inherit biases embedded within historical data, particularly when observable variables are used as proxies for unobserved clinical need. Similar concerns arise in mental health triage systems, where referral documentation and patterns of healthcare access may influence model predictions.

### 2.3 CHRONOSIG’s Fairness Approach and its Limitations

CHRONOSIG acknowledges the potential for algorithmic bias and highlights the importance of distributive justice principles in clinical AI development [Joyce et al., 2021]. However, its fairness approach focuses primarily on identifying risks within the development process rather than providing a formal framework for threshold selection.

In particular, CHRONOSIG does not define a method for quantifying fairness gaps, incorporating demographic differences in triage error costs, or evaluating formal criteria such as equalised odds. Additionally, Joyce et al. (2021) note the use of a heuristic proxy for triage outcomes, where referrals are classified as rejected if no clinical notes are entered within 14 days, potentially introducing uncertainty into the training labels.

This project addresses this gap by applying fairness theory to threshold selection, using Chouldechova’s impossibility result and a Bayesian decision framework to investigate how demographic trade-offs can be managed through group-specific thresholds.

## 3 Methodology

### 3.1 Formalising the Triage Decision

I model an AI-assisted mental health triage system, motivated by CHRONOSIG, as a binary classifier operating on a continuous risk score  $s \in [0, 1]$ . This score represents the model’s estimated likelihood that a patient requires escalation to secondary mental health care. A decision threshold  $\tau \in [0, 1]$  is applied to convert this continuous score into a binary triage decision  $D \in \{0, 1\}$ :

$$D = \begin{cases} 1 & \text{if } s \geq \tau \\ 0 & \text{if } s < \tau \end{cases} \quad (5)$$

where  $D = 1$  denotes referral to secondary mental health care and  $D = 0$  denotes non-referral. The threshold  $\tau$  is applied uniformly across all patients, regardless of demographic group.

For each patient, the true clinical status  $T \in \{0, 1\}$  represents the ground truth outcome, independent of the model’s prediction, where  $T = 1$  indicates genuine clinical need for secondary care. The resulting classification outcomes are:

- **True Positive (TP):**  $D = 1, T = 1$  — correctly referred
- **False Positive (FP):**  $D = 1, T = 0$  — referred despite no clinical need
- **True Negative (TN):**  $D = 0, T = 0$  — correctly not referred
- **False Negative (FN):**  $D = 0, T = 1$  — not referred despite clinical need

The true positive rate (TPR) and false positive rate (FPR) for a demographic group  $G$  at threshold  $\tau$  are defined as:

$$\text{TPR}_G(\tau) = P(D = 1 \mid T = 1, G) \quad (6)$$

$$\text{FPR}_G(\tau) = P(D = 1 \mid T = 0, G) \quad (7)$$

The base rate for demographic group  $G$  is defined as:

$$P_G = P(T = 1 \mid G) \quad (8)$$

This represents the proportion of patients in group  $G$  who have a true need for secondary care. Differences in representation, access to care, and historical clinical processes may lead to differing base rates between groups, such that  $P_A \neq P_B$ .

### 3.2 Bayesian Loss Framework

Classification errors in triage decisions take two forms: a false negative, where a patient requiring secondary care is not referred, and a false positive, where a patient is referred despite not requiring secondary care. These errors have different clinical and societal consequences and should therefore not be treated as equivalent.

I formalise this decision-theoretic approach by defining an asymmetric loss function  $L(D, T)$ , which assigns explicit costs to different classification outcomes. Let  $c_{\text{FN}}$  represent the cost of a false negative error (failing to refer a patient who requires secondary care), and let  $c_{\text{FP}}$  represent the cost of a false positive error (referring a patient who does not require secondary care).

The expected loss  $\mathbb{E}[L]$  across a clinical population is:

$$\mathbb{E}[L] = c_{\text{FN}} \cdot P(\text{FN}) + c_{\text{FP}} \cdot P(\text{FP}) \quad (9)$$

To minimise this expected loss for an individual patient, a referral should be made ( $D = 1$ ) if the expected cost of not referring exceeds the expected cost of referral. Given a patient's risk score  $s$ , the optimal decision rule therefore refers when:

$$c_{\text{FN}} \cdot P(T = 1 \mid s) > c_{\text{FP}} \cdot P(T = 0 \mid s) \quad (10)$$

Rearranging gives the relationship between risk and error costs:

$$\frac{P(T = 1 \mid s)}{P(T = 0 \mid s)} > \frac{c_{\text{FP}}}{c_{\text{FN}}} \quad (11)$$

This demonstrates that the optimal threshold depends on the ratio of error costs. Importantly, these costs may differ between demographic groups. For example, the consequences of a missed referral may be greater for individuals with fewer alternative sources of support or reduced access to care. A uniform threshold does not account for potential differences in these error costs between groups.

### 3.3 Deriving the Fairness Impossibility Result

We now derive the relationship between calibration and equalised odds, demonstrating why these fairness criteria cannot both be satisfied when demographic groups have different base rates.

#### Step 1: Expressing PPV via Bayes' Theorem.

The positive predictive value for group  $G$  is defined as:

$$\text{PPV}_G = P(T = 1 \mid D = 1, G) \quad (12)$$

Applying Bayes' theorem:

$$\text{PPV}_G = \frac{P(D = 1 \mid T = 1, G) \cdot P(T = 1 \mid G)}{P(D = 1 \mid G)} \quad (13)$$

Using the law of total probability, the denominator can be expanded as:

$$P(D = 1 \mid G) = \text{TPR}_G \cdot P_G + \text{FPR}_G \cdot (1 - P_G) \quad (14)$$

Substituting  $\text{TPR}_G = 1 - \text{FNR}_G$ , we obtain:

$$\text{PPV}_G = \frac{(1 - \text{FNR}_G) \cdot P_G}{(1 - \text{FNR}_G) \cdot P_G + \text{FPR}_G \cdot (1 - P_G)} \quad (15)$$

#### Step 2: The Impossibility Result.

Assume equalised odds is satisfied, such that:

$$\text{FNR}_A = \text{FNR}_B = \text{FNR}, \quad \text{FPR}_A = \text{FPR}_B = \text{FPR} \quad (16)$$

For calibration to also hold, the positive predictive values must be equal:

$$\text{PPV}_A = \text{PPV}_B \quad (17)$$

Substituting Equation 15 for both groups and cross-multiplying gives:

$$(1 - \text{FNR})P_A[(1 - \text{FNR})P_B + \text{FPR}(1 - P_B)] = (1 - \text{FNR})P_B[(1 - \text{FNR})P_A + \text{FPR}(1 - P_A)] \quad (18)$$

Expanding and cancelling the common term  $(1 - \text{FNR})^2 P_A P_B$  from both sides gives:

$$(1 - \text{FNR})\text{FPR}P_A(1 - P_B) = (1 - \text{FNR})\text{FPR}P_B(1 - P_A) \quad (19)$$

For a non-trivial classifier, dividing by  $(1 - \text{FNR})\text{FPR}$  gives:

$$P_A(1 - P_B) = P_B(1 - P_A) \quad (20)$$

which simplifies to:

$$P_A = P_B \quad (21)$$

This contradicts the assumption that demographic groups have different base rates ( $P_A \neq P_B$ ). Therefore, when base rates differ between groups, equalised odds and calibration cannot be simultaneously satisfied by a non-trivial classifier.

#### **Numerical Confirmation.**

Using the base rates specified in our simulation ( $P_A = 0.293$ ,  $P_B = 0.496$ ), and assuming equal error rates ( $\text{FNR} = \text{FPR} = 0.1$ ):

$$\text{PPV}_A = \frac{0.9 \times 0.293}{0.9 \times 0.293 + 0.1 \times 0.707} = 0.789 \quad (22)$$

$$\text{PPV}_B = \frac{0.9 \times 0.496}{0.9 \times 0.496 + 0.1 \times 0.504} = 0.899 \quad (23)$$

The resulting PPV gap of  $0.899 - 0.789 = 0.110$  remains even when equal error rates are assumed. The simulation further demonstrates that the PPV gap remains non-zero across the evaluated range of FNR and FPR values, consistent with the theoretical result that calibration and equalised odds cannot both be achieved when base rates differ.

### 3.4 Group-Specific ROC Thresholds

Given the impossibility of simultaneously satisfying all fairness criteria under a uniform threshold, I propose a decoupled decision framework that aims to improve alignment with equalised odds while making the resulting trade-offs explicit.

**ROC Curves.** For each demographic group  $G$ , a Receiver Operating Characteristic (ROC) curve represents the achievable (FPR, TPR) pairs as the decision threshold  $\tau$  varies over  $[0, 1]$ :

$$\text{ROC}_G = \{(\text{FPR}_G(\tau), \text{TPR}_G(\tau)) : \tau \in [0, 1]\} \quad (24)$$

When model performance differs between demographic groups, their ROC curves may differ. A uniform threshold  $\tau$  selects the same decision boundary for all groups, which can result in different TPR and FPR values across groups. Therefore, equalised odds cannot generally be achieved without allowing thresholds to vary between demographic groups.

**Optimal Group-Specific Thresholds.** We select separate thresholds  $\tau_A^*$  and  $\tau_B^*$  for each group by minimising group-specific expected Bayesian loss:

$$\tau_G^* = \arg \min_{\tau \in [0, 1]} [c_{\text{FN}}^G \cdot \text{FNR}_G(\tau) \cdot P_G + c_{\text{FP}}^G \cdot \text{FPR}_G(\tau) \cdot (1 - P_G)] \quad (25)$$

where  $c_{\text{FN}}^G$  and  $c_{\text{FP}}^G$  represent the costs of false negative and false positive errors for group  $G$ . Different optimal thresholds may arise due to differences in error costs, base rates, or model performance between groups.

**Cost Parameters.** We set  $c_{\text{FN}} = 5.0$  and  $c_{\text{FP}} = 1.0$  as a modelling assumption, reflecting that failing to refer a patient who requires mental health support is considered more costly than an unnecessary referral. Under this cost ratio, the optimal thresholds obtained were  $\tau_A^* = 0.353$  and  $\tau_B^* = 0.439$  under Grid Search tuning, and  $\tau_A^* = 0.404$  and  $\tau_B^* = 0.446$  under Bayesian Optimisation.

**Two-Stage Bias Reduction.** The proposed framework operates in two stages. Stage 1 investigates whether model configuration can influence reliance on potential proxy variables. Hyperparameter optimisation is applied to evaluate whether different model settings reduce reliance on features such as referral letter length, which may capture documentation patterns associated with demographic differences rather than clinical need alone. Stage 2 applies group-specific ROC thresholds to address remaining disparities arising from differences in model performance and base rates. Chouldechova’s result demonstrates that reducing model-level bias alone cannot guarantee equal fairness outcomes when  $P_A \neq P_B$ ; therefore, threshold selection remains a necessary component of fairness-aware decision-making. This structure is consistent with prior work on decoupled classifiers [Dwork et al., 2018].

### 3.5 A Theorem on Optimal Group-Specific Thresholds

The previous sections have established that group-specific thresholds are necessary to correct the fairness gap identified by Chouldechova’s impossibility result. We now formalise the conditions under which group-specific thresholds are not merely fair but *optimal* under asymmetric Bayesian loss.

**Theorem 1** (Optimal Group-Specific Thresholds). *Let  $G \in \{A, B\}$  denote two demographic groups with base rate  $P_G = P(T = 1 \mid G)$ , group-specific error costs  $c_{FN}^G, c_{FP}^G > 0$ , and differentiable ROC curves  $ROC_G$ . Define the optimal threshold as:*

$$\tau_G^* = \arg \min_{\tau \in [0,1]} [c_{FN}^G \cdot FNR_G(\tau) \cdot P_G + c_{FP}^G \cdot FPR_G(\tau) \cdot (1 - P_G)] \quad (26)$$

Assuming an interior optimum, the first-order condition at  $\tau_G^*$  is:

$$\left. \frac{d \text{TPR}_G}{d \text{FPR}_G} \right|_{\tau_G^*} = \frac{c_{FP}^G(1 - P_G)}{c_{FN}^G P_G} \quad (27)$$

Therefore, the optimal threshold corresponds to the point on group  $G$ ’s ROC curve where the slope equals the cost-adjusted base-rate ratio.

*Proof.* The expected loss for group  $G$  is:

$$\mathcal{L}_G(\tau) = c_{FN}^G \cdot \text{FNR}_G(\tau) P_G + c_{FP}^G \cdot \text{FPR}_G(\tau) (1 - P_G)$$

Differentiating with respect to  $\tau$  and setting the derivative to zero gives:

$$-c_{FN}^G P_G \frac{d \text{TPR}_G}{d \tau} + c_{FP}^G (1 - P_G) \frac{d \text{FPR}_G}{d \tau} = 0 \quad (28)$$

Rearranging and dividing by  $\frac{d \text{FPR}_G}{d \tau}$  gives Equation 27.  $\square$

**Corollary 1.** *When either  $P_A \neq P_B$  or the relative error costs differ between groups, the optimal ROC slope condition differs between groups. Consequently, the optimal thresholds will generally differ. Therefore, a uniform threshold may be suboptimal for at least one group under asymmetric loss. Group-specific thresholds provide a decision-theoretic approach for selecting thresholds that account for group-specific characteristics.*

This result connects the three components of this project. The Bayesian loss framework determines the cost-adjusted ROC slope required for an optimal decision. Chouldechova’s result demonstrates why differing base rates create unavoidable fairness trade-offs. The ROC framework provides the geometric

interpretation of threshold selection. Together, these results motivate group-specific threshold optimisation as a principled decision-theoretic approach rather than an ad hoc fairness adjustment.

### 3.6 Simulation Design

**Synthetic Data Generation.** Due to restrictions on access to NHS Electronic Health Records, this project uses synthetic patient data informed by documented patterns of demographic variation in NHS mental health service usage [Joyce et al., 2021, NHS Digital, 2024]. The purpose of this simulation is not to reproduce an NHS population exactly, but to evaluate the fairness behaviour of AI-assisted triage systems under controlled conditions.

For the CHRONOSIG-inspired simulation, each synthetic patient is represented using clinical and documentation-related features, including risk score, referral letter length, number of previous contacts with services, and age. Two simulated demographic groups are defined:

- **Group A** (simulated underrepresented group,  $n = 200$ ): risk scores drawn from  $\text{Beta}(1.5, 4.0)$ , referral letter length from  $\mathcal{N}(150, 50^2)$ , and clinical need base rate  $P_A = 0.30$ .
- **Group B** (simulated majority group,  $n = 800$ ): risk scores drawn from  $\text{Beta}(2.5, 3.0)$ , referral letter length from  $\mathcal{N}(300, 80^2)$ , and clinical need base rate  $P_B = 0.50$ .

The Beta distribution is the natural choice for probability scores  $s \in [0, 1]$ . Different parameters for each group reflect the documented pattern whereby underrepresented patients produce shorter, vaguer referral letters due to structural barriers to healthcare access [Obermeyer et al., 2019].

**Classifier.** A Random Forest classifier is selected as the primary model for three reasons: it handles mixed data types without normalisation, its bagging mechanism reduces overfitting to majority group patterns, and its feature importance scores provide interpretable evidence of where bias enters the model. Crucially, while bagging reduces variance, it cannot remove systematic bias from base rate differences in training data — motivating the need for Stage 2 threshold correction.

**Permutation Importance.** Feature bias is quantified using the permutation importance formula:

$$I_j = \frac{1}{T} \sum_{t=1}^T (E_t^{(j)} - E_t) \quad (29)$$

where  $T$  is the number of trees,  $E_t$  is the classification error of tree  $t$ , and  $E_t^{(j)}$  is the error after randomly permuting feature  $j$ . A positive value of  $I_j$  indicates

the model relies on feature  $j$ ; a negative value indicates the feature introduces noise.

**Hyperparameter Optimisation.** Two strategies are compared for Stage 1 bias reduction:

1. **Grid Search:** exhaustive evaluation of 48 valid parameter combinations over  $\text{max\_features} \in \{1, 2, 3, \sqrt{p}\}$ ,  $\text{max\_samples} \in \{0.5, 0.6, 0.7, 1.0\}$ ,  $\text{bootstrap} \in \{\text{True}, \text{False}\}$ ,  $\text{max\_depth} \in \{3, 5, 7\}$ .
2. **Bayesian Optimisation:** 50 evaluations guided by a Gaussian Process surrogate model over a continuous parameter space, enabling exploration between discrete grid points.

**SHAP Analysis.** SHAP values [Lundberg and Lee, 2017] are computed using TreeExplainer to provide individual-level prediction explanations. Unlike permutation importance, SHAP values reveal the direction and magnitude of each feature’s contribution — identifying the mechanism of bias rather than merely its existence.

**Model Benchmark.** To confirm the fairness gap is structural rather than architecture-specific, the simulation is replicated using XGBoost and LightGBM. If all three architectures exhibit comparable TPR gaps under a uniform threshold, this empirically supports Chouldechova’s claim that bias arises from base rate differences rather than model choice.

**CAMHS Extension.** The framework is extended to Child and Adolescent Mental Health Services using synthetic data generated from two validated clinical questionnaires: the Strengths and Difficulties Questionnaire (SDQ, 25 items, 3 response options) and the Revised Child Anxiety and Depression Scale (RCADS, 47 items, 4 response options). Both parent and child perspectives are modelled. An engineered feature, `underreporting_index`, captures the tendency of young carers to minimise reported difficulties and serves as the CAMHS equivalent of referral letter length in the CHRONOSIG simulation.

## 4 Results

### 4.1 Baseline Fairness Gap

Under the baseline Random Forest classifier with default parameters and a uniform decision threshold  $\tau = 0.4$ , the simulation reveals a substantial disparity in true positive rates between demographic groups. Group A (the simulated underrepresented group) achieves a true positive rate of  $\text{TPR}_A = 0.353$ , compared with  $\text{TPR}_B = 0.933$  for Group B, producing a TPR gap of 0.580.

This corresponds to a false negative rate of 64.7% for Group A, compared with 6.7% for Group B. Therefore, under a uniform threshold, positive cases within Group A are substantially more likely to be missed than positive cases within

Group B, demonstrating that a single decision threshold can produce unequal error rates across demographic groups.

The observed base rates in the simulation are  $P_A = 0.293$  and  $P_B = 0.496$ . These unequal prevalence rates represent the condition under which Chouldechova’s impossibility result highlights the difficulty of simultaneously achieving calibration and equalised error rates. Under the uniform threshold, the evaluated fairness criteria show substantial disparities between groups:

$$I_j = \frac{1}{T} \sum_{t=1}^T (E_t^{(j)} - E_t) \quad (30)$$

produces an importance value of  $I_{\text{referral\_length}} = 0.0242$ , further indicating that referral length contributes to the model’s predictive behaviour.

SHAP analysis provides insight into the direction of this feature contribution. The mean SHAP value for referral letter length is  $-0.0761$  for Group A patients compared with  $+0.0092$  for Group B patients, producing a difference of  $-0.0853$ . This indicates that referral length contributes more negatively to model predictions for Group A patients than for Group B patients. Furthermore, 94.8% of Group A patients have negative SHAP values for this feature, suggesting that the relationship between referral length and model predictions is systematic rather than random.

Group-specific permutation importance further demonstrates this asymmetry: referral letter length has an importance value of 0.0155 for Group A compared with 0.0050 for Group B. This indicates that referral length has a greater influence on predictions for Group A than Group B, suggesting that documentation-related features may contribute to the observed disparity in model behaviour.

A critical finding emerges from this comparison: Bayesian Optimisation identifies parameter values — specifically `max_samples=0.734` — that lie between the discrete values explored by Grid Search (0.6 and 0.7). This demonstrates the benefit of continuous parameter exploration and highlights its ability to identify parameter configurations associated with improved fairness outcomes.

## 4.2 Stage 2: Group-Specific Threshold Correction

For the Grid Search tuned model, applying group-specific thresholds  $\tau_A^* = 0.353$  and  $\tau_B^* = 0.439$  selected using the Bayesian expected loss function (Equation 25) eliminates the TPR gap entirely:

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Under group-specific thresholds, three of the four evaluated fairness criteria are satisfied simultaneously: demographic parity (1.000 vs 1.000), equalised odds TPR (1.000 vs 1.000), and equalised odds FPR (1.000 vs 1.000). Calibration

remains violated ( $PPV_A = 0.293$ ,  $PPV_B = 0.496$ ), consistent with Chouldechova’s impossibility result: when  $P_A \neq P_B$ , calibration cannot generally be achieved simultaneously with equalised odds for classifiers with imperfect prediction performance.

However, this improvement in fairness metrics should be interpreted alongside predictive behaviour, as equalising error rates through threshold adjustment may increase the number of positive predictions. Therefore, fairness improvements must be considered alongside clinical utility and model performance.

### 4.3 Numerical Evaluation of Chouldechova’s Result

The simulation provides numerical evidence consistent with the impossibility result described in Section 3.3. Assuming hypothetical equal error rates ( $FNR = FPR = 0.1$ ) and using the observed base rates ( $P_A = 0.293$ ,  $P_B = 0.496$ ), Chouldechova’s formula produces  $PPV_A = 0.789$  and  $PPV_B = 0.899$ , resulting in a PPV disparity of 0.110. This difference remains despite equalised odds being perfectly satisfied, demonstrating the tension between calibration and equalised error rates under unequal base rates.

A systematic sweep across the evaluated combinations of  $FNR \in [0.05, 0.50]$  and  $FPR \in [0.05, 0.50]$  further demonstrates this limitation. The minimum achievable PPV gap within the evaluated parameter space is 0.0619, which remains greater than zero. These findings provide numerical support for Chouldechova’s result, illustrating that calibration and equalised odds cannot generally be achieved simultaneously when groups have different base rates.

### 4.4 Model Architecture Benchmark

To examine whether the observed fairness disparity is specific to the Random Forest architecture, the simulation is replicated using XGBoost and LightGBM classifiers under identical experimental conditions.

| Model                    | TPR A | TPR B | Gap (Uniform) | Gap (Optimal) |
|--------------------------|-------|-------|---------------|---------------|
| Random Forest (Baseline) | 0.353 | 0.933 | 0.580         | 0.000         |
| Random Forest (Bayesian) | 1.000 | 1.000 | 0.000         | 0.000         |
| XGBoost                  | 0.412 | 0.667 | 0.255         | 0.000         |
| LightGBM                 | 0.412 | 0.650 | 0.238         | 0.000         |

Table 1: Model benchmark: TPR disparity across architectures under uniform and optimised threshold configurations.

All three model architectures exhibit a positive TPR disparity under the uniform threshold, suggesting that the observed fairness issue is not specific to a single classifier architecture. Under the optimised threshold configurations, all evaluated architectures achieve a measured TPR gap of 0.000, indicating equal group-level true positive rates according to this fairness criterion.

However, equality in TPR alone does not imply overall model optimality. The resulting threshold configurations must also be interpreted alongside other performance measures, including false positive rates, calibration, and prediction distributions. Therefore, the observed reduction in TPR disparity demonstrates improvement according to the selected fairness metric, rather than a complete resolution of all fairness and utility trade-offs.

## 4.5 CAMHS Extension

The proposed framework is further evaluated using a CAMHS simulation based on synthetic SDQ and RCADS questionnaire data. In this setting, Group A patients represent young carers, where simulated underreporting behaviour (with an underreporting factor of 0.35) and missing parent data (with a 25% missing rate) produce lower observed questionnaire scores despite higher simulated underlying clinical need.

Under the baseline Random Forest classifier with a uniform threshold, the CAMHS simulation produces a TPR gap of 0.212 ( $\text{TPR}_A = 0.759$ ,  $\text{TPR}_B = 0.971$ ). The most influential feature associated with this disparity is **underreporting\_index**, with a feature importance value of 0.0806. This feature demonstrates a similar role to referral letter length in CHRONOSIG by capturing differences in observed reporting behaviour.

Chouldechova’s impossibility result also applies within the CAMHS simulation, where base rates of  $P_A = 0.460$  and  $P_B = 0.582$  satisfy the unequal prevalence condition. The architecture comparison indicates that fairness disparities persist across model choices, with TPR gaps of 0.212, 0.191, and 0.327 for Random Forest, XGBoost, and LightGBM respectively. Group-specific threshold correction reduces the measured TPR gap to 0.000 for each architecture.

This extension demonstrates that the proposed framework is not limited to CHRONOSIG or adult secondary care. Instead, it provides a general approach for analysing and mitigating fairness disparities in mental health AI triage systems operating across populations with different base rates.

# 5 Discussion

## 5.1 The Structural Nature of Algorithmic Bias

The central finding of this project is that fairness disparities in NHS mental health AI triage can arise from the structure of the prediction problem itself, rather than solely from deficiencies in model architecture or implementation. Chouldechova’s impossibility result, derived in Section 3.3, demonstrates that when demographic groups have different base rates of clinical outcomes, calibration, equalised odds, and demographic parity cannot generally be achieved simultaneously by a single decision threshold.

This finding has important implications for NHS AI governance. Current approaches, including CHRONOSIG’s stated commitment to distributive justice principles, emphasise procedural safeguards such as clinician oversight and monitoring of demographic performance. While these measures remain important, they cannot fully resolve mathematical trade-offs that emerge from unequal base rates and differing patterns of observed data. The results of this study suggest that fairness interventions must therefore consider not only model development practices, but also the underlying data generation process and decision thresholds through which predictions are converted into clinical actions.

## 5.2 Proxy Variables as a Mechanism of Bias

SHAP and permutation importance analyses suggest that referral letter length acts as a potential proxy feature within the CHRONOSIG simulation. Referral length demonstrates a substantially different contribution pattern across demographic groups, with negative SHAP contributions observed for 94.8% of Group A patients. This indicates that documentation-related features may influence predictions differently across groups and contribute to observed disparities in model behaviour.

This finding extends Obermeyer et al.’s (2019) observation that healthcare algorithms may learn patterns associated with historical healthcare processes rather than underlying clinical need. In the CHRONOSIG context, referral documentation characteristics may capture differences in healthcare access, communication barriers, or referral practices rather than differences in clinical severity. As a result, features related to documentation quality may introduce unequal predictive effects when used by automated decision systems.

The CAMHS extension demonstrates that similar mechanisms may occur across different care settings. In the CAMHS simulation, the equivalent feature is `underreporting_index`, which captures differences in reported difficulties among young carers. Although the specific feature differs from referral length, both scenarios demonstrate a similar pattern: variables associated with differences in observed data generation processes can influence model predictions in ways that contribute to group disparities. This highlights the importance of examining feature behaviour and data-generating mechanisms when evaluating fairness in healthcare AI systems.

## 5.3 The Two-Stage Framework as a Practical Response

The proposed two-stage framework provides a practical approach for reducing fairness disparities within an otherwise constrained optimisation problem. Stage 1 hyperparameter tuning reduces the measured influence of referral letter length, demonstrated by a 60% reduction in referral length permutation importance under Grid Search and an 87% reduction under Bayesian Optimisation. Bayesian Optimisation achieves further improvement by exploring continuous parameter values beyond the discrete configurations evaluated by Grid Search, highlighting

the potential benefit of continuous parameter exploration when seeking models with improved fairness characteristics.

However, Stage 1 alone does not fully address the fairness disparity. While hyperparameter optimisation can alter model behaviour and reduce dependence on specific features, differences in base rates ( $P_A \neq P_B$ ) remain a property of the underlying data distribution. As demonstrated by Chouldechova’s result, such differences create fundamental trade-offs between fairness criteria. Stage 2 group-specific threshold correction addresses the remaining disparity directly by adjusting decision boundaries between groups, achieving a measured TPR gap = 0.000 across all evaluated model architectures.

Importantly, the framework does not attempt to overcome the impossibility result but instead makes an explicit fairness trade-off. By prioritising equalised odds over calibration, the approach reflects a decision that equal opportunity for identifying patients requiring support may be more clinically valuable than requiring predicted probabilities to have identical interpretations across groups. This represents a normative design choice informed by the potential consequences of unequal access to necessary care.

## 5.4 Comparison with Existing Approaches

The proposed framework relates to and extends existing approaches in algorithmic fairness. The decoupled classifier approach of Dwork et al. [2018] proposes training separate models for different demographic groups; in contrast, this project maintains a shared predictive model while applying group-specific thresholds, preserving a unified risk estimation framework while addressing threshold-level disparities. The Bayesian loss formulation further extends standard threshold optimisation by incorporating asymmetric consequences associated with triage errors, allowing threshold selection to reflect clinically motivated cost considerations.

The model benchmark across Random Forest, XGBoost, and LightGBM indicates that the proposed threshold correction approach can be applied across multiple classifier architectures. This model flexibility is important for potential deployment settings, where different NHS organisations may adopt different underlying machine learning approaches.

## 5.5 Limitations

Several important limitations should be acknowledged. First, the simulation uses synthetic data designed to represent plausible demographic disparities rather than real NHS patient records. Although simulation parameters are informed by published evidence, the results should be interpreted as demonstrating methodological behaviour rather than providing definitive estimates of real-world fairness performance. A rigorous evaluation using real CHRONOSIG data would

require appropriate ethical approvals, governance processes, and access to suitable NHS datasets.

Second, the binary demographic grouping ( $G \in \{A, B\}$ ) represents a simplification of real-world populations. Individuals may possess multiple intersecting characteristics, including ethnicity, gender, socioeconomic status, and caring responsibilities, which can interact to produce more complex patterns of disadvantage. Future work should extend the framework to multi-group settings, potentially incorporating objectives such as minimising worst-group error rates.

Third, the cost parameters  $c_{FN} = 5.0$  and  $c_{FP} = 1.0$  are specified heuristically based on clinical reasoning rather than empirical measurement. Different cost assumptions will produce different optimal thresholds. A real-world deployment would therefore require engagement with clinicians, commissioners, and patient representatives to establish cost parameters that reflect agreed priorities and values.

Fourth, counterfactual fairness — which considers whether a model decision would change if an individual’s demographic characteristics were altered while holding relevant factors constant — is not addressed in this framework. This stronger form of fairness, formalised through causal approaches such as Pearl’s structural causal models [Kusner et al., 2017], would require a detailed causal representation of the referral pathway and remains an important area for future research.

## 5.6 Future Directions

The framework developed in this project suggests several directions for future research. One potential extension is application to NHS Talking Therapies (formerly IAPT), where machine learning systems such as Limbic and similar tools support mental health referral processes using questionnaire-based information including PHQ-9 and GAD-7 measures. While the specific clinical setting and proxy features may differ, these systems provide another opportunity to investigate how unequal base rates and observed data patterns influence algorithmic fairness.

A second direction is longitudinal modelling. Duration and progression of difficulties are clinically important predictors of mental health outcomes, yet many current triage approaches treat referrals as independent events. Incorporating temporal information into fairness analysis could provide further insight into how delayed recognition and historical barriers affect model predictions.

Finally, future work could explore translation of the proposed framework into practical NHS guidance for fairness-aware threshold selection and model evaluation. Developing deployment-oriented recommendations would represent an important step towards applying the mathematical insights developed in this project within real clinical decision-support environments.

## 6 Conclusion

This project has demonstrated that fairness disparities in NHS mental health AI triage can arise from the interaction between unequal base rates, observed data patterns, and the use of uniform decision thresholds. Through a derivation from first principles, this study has shown how Chouldechova’s (2017) impossibility result applies to the CHRONOSIG simulation: when groups have different base rates ( $P_A \neq P_B$ ), calibration and equalised odds cannot generally be achieved simultaneously using a single threshold.

The simulation provides empirical evidence of this theoretical limitation. Under the baseline model, Group A patients experience a false negative rate of 64.7% compared with 6.7% for Group B, producing a substantial disparity in missed positive cases. Feature analysis identifies referral letter length as a potential proxy feature associated with this disparity. SHAP analysis demonstrates that 94.8% of Group A patients have negative SHAP contributions for referral length, suggesting that documentation-related features influence model predictions differently across demographic groups.

The proposed two-stage framework addresses this disparity through a combination of hyperparameter optimisation and group-specific threshold correction. Stage 1 reduces the measured influence of referral length within the model, while Stage 2 adjusts decision thresholds to reduce unequal error rates between groups. Across the evaluated model architectures — Random Forest, Bayesian-optimised Random Forest, XGBoost, and LightGBM — threshold correction reduces the measured TPR disparity to 0.000 under the evaluated fairness metric. This result should be interpreted alongside broader predictive utility measures. Extension to a CAMHS simulation using SDQ and RCADS questionnaire data further demonstrates that similar fairness mechanisms can emerge in different mental health contexts, with underreporting behaviour acting as an analogous source of unequal predictive influence.

Importantly, this framework does not attempt to overcome Chouldechova’s impossibility result; instead, it makes the associated fairness trade-offs explicit. By replacing an implicit uniform threshold with a mathematically justified threshold selection process, the framework enables NHS commissioners, clinicians, and AI developers to evaluate which fairness objectives should be prioritised in a given clinical context. In mental health triage, where delayed recognition of patients requiring support may carry significant consequences, prioritising equal opportunity of identification represents a defensible ethical and clinical choice.

The motivation for this project is closely connected to its research objective. As a young carer with experience of NHS mental health and social care systems, I have seen how barriers in healthcare pathways can affect whether individuals receive appropriate support. This project seeks to make such disparities measurable by combining mathematical analysis, fairness evaluation, and practical mitigation strategies.

NHS mental health AI systems increasingly influence decisions about access to care. Every threshold choice represents a trade-off between competing objectives, and those trade-offs should be visible, measurable, and open to scrutiny. Ensuring that fairness considerations are incorporated into the design and evaluation of these systems is therefore not only a technical challenge, but an important requirement for responsible clinical AI deployment.

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